

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3371

3 Credits: 3

Program Core

Source-grounded

Introduction to IC Fabrication

- To introduce the fundamentals of IC fabrication process.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3371 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3371.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	3371
Course title	Introduction to IC Fabrication
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:3 T:0 P:0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the fundamentals of IC fabrication process.

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding preparation and wafer cleaning.	See official syllabus
CO2	Explain the process of epitaxy and oxidation. 11 Understanding	See official syllabus
CO3	Explain lithography and etching process. 10 Understanding Illustrate the process of deposition, diffusion, ion	See official syllabus
CO4	12 Understanding implantation and packaging. Series Exam 2	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	cleaning	4	As specified
Module 2	Explain the process of Epitaxy and Oxidation.	4	As specified
Module 3	Explain Lithography and Etching process.	4	As specified
Module 4	Packaging.	4	As specified

MODULE 1**cleaning**

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cleaning
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding fabrication

2 Understanding fabrication is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding fabrication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding fabrication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding fabrication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding fabrication

പ്രദേശം definition/meaning. പദങ്ങൾ, steps, application. Technical terms English-ൽ പദങ്ങൾ.

▼ Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing

Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the methods of single crystal growth 3 understanding explain chemical-mechanical polishing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing
 ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം
 ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms
 English- ഫലപ്രസാദം ഫലപ്രസാദം.

▼ 2 Understanding technique

2 Understanding technique is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding technique using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding technique should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding technique means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding technique

പ്രദേശം definition/meaning പ്രദേശം. പ്രദേശം പ്രദേശം പ്രദേശം, steps, application പ്രദേശം പ്രദേശം. Technical terms English- പ്രദേശം പ്രദേശം.

▼ **Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.**

Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain wafer preparation 3 understanding crystal growth, wafer preparation: introduction & historical perspective- clean room concept- growth of single crystal si-czochralski(cz) and float zone(fz) processes. surface contamination-chemical-mechanical polishing. wafer preparation-di water, chemical cleaning-rca, processing consideration- getting the thermal stress factor etc. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain wafer preparation 3

Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English- പദപരിചയം പദപരിചയം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the process of Epitaxy and Oxidation.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the process of Epitaxy and Oxidation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth

Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the process of epitaxy 3 understanding illustrate epitaxial valuation and growth should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth definition/meaning . . . , steps, application Technical terms English-

▼ 3 Understanding mechanism

3 Understanding mechanism is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mechanism means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding mechanism

മെക്കാനിസം എന്നതിന്റെ definition/meaning നൽകുക. മെക്കാനിസം എങ്ങനെ പ്രവർത്തിക്കുന്നു, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

▼ Outline the process of oxidation

Outline the process of oxidation is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the process of oxidation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the process of oxidation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the process of oxidation means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓക്സീകരണം Outline the process of oxidation
 ഓക്സീകരണം എന്നതിന്റെ definition/meaning എന്താണ്. ഓക്സീകരണം
 ഏതു ഘട്ടങ്ങളിലൂടെ നടക്കുന്നു, steps, application ഏതെല്ലാം ഓക്സീകരണങ്ങൾ ഉണ്ട്. Technical terms
 English-ൽ എങ്ങനെ എഴുതണം.

▼ Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.

Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the kinetics of oxidation 2 understanding epitaxy and oxidation: epitaxy process-vapor phase epitaxy, molecular beam epitaxy. epitaxial valuation-growth mechanism and kinetics. oxidation techniques-dry and wet oxidation. device isolation-locos-kinetics of oxidation. deal grove's model-oxidation system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and

kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.

പ്രതിരോധനം പ്രതിരോധനം definition/meaning പ്രതിരോധനം പ്രതിരോധനം പ്രതിരോധനം, steps, application പ്രതിരോധനം പ്രതിരോധനം പ്രതിരോധനം. Technical terms English-പ്രതിരോധനം പ്രതിരോധനം പ്രതിരോധനം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain Lithography and Etching process.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Lithography and Etching process.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Illustrate the concept of Lithography

Illustrate the concept of Lithography is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concept of Lithography using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concept of lithography should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concept of Lithography means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓ ഓരോന്നും Lithography എന്നതിന്റെ definition/meaning എഴുതുക. ഓരോന്നും ഓരോന്നും steps, application എന്നിവ എഴുതുക. Technical terms English-ഓ എഴുതുക.

▼ Explain different types of Lithographic processes

Explain different types of Lithographic processes is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different types of Lithographic processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of lithographic processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different types of Lithographic processes means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain different types of Lithographic processes. definition/meaning. steps, application. Technical terms English-.

▼ Explain the concept of etching

Explain the concept of etching is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of etching using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of etching should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of etching means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു പറയുന്നതാണ് etching എന്നതിന്റെ definition/meaning എന്താണ്. എന്താണ് etching എന്നതിന്റെ steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ എന്താണ് എന്നും പറയണം.

▼ Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and

Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma

Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the different types of etching process 3 understanding lithography & etching: lithography-overview of lithography-radiation sources-masks-photoresist-components of photoresist- optical aligners-resolution-depth of focus-advanced lithography-electron beam lithography-x-ray lithography-ion lithography. etching-anisotropy-selectivity-wet etching, plasma etching, plasma properties-reactive - ion etching. illustrate the process of deposition, diffusion, ion implantation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Packaging.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Packaging.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain Diffusion mechanism

Explain Diffusion mechanism is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Diffusion mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain diffusion mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Diffusion mechanism means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain Diffusion mechanism

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Explain various ion implantation techniques

Explain various ion implantation techniques is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various ion implantation techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various ion implantation techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various ion implantation techniques means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഐ Explain various ion implantation techniques ഐഐഐഐഐഐഐഐ definition/meaning ഐഐഐഐഐഐഐഐ. ഐഐഐഐ ഐഐഐഐ, steps, application ഐഐഐഐ ഐഐഐഐഐഐഐഐ. Technical terms English-ഐ ഐഐഐ ഐഐഐഐഐഐഐഐ.

▼ Explain different deposition mechanisms

Explain different deposition mechanisms is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different deposition mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different deposition mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different deposition mechanisms means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain different deposition mechanisms
 പദപരിചയം definition/meaning. പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms
 English- പദപരിചയം പദപരിചയം.

▼ Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory- Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in>

Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated

purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory- Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain metallization and packaging 3 understanding deposition, diffusion, ion implantation and packaging: model of diffusion in solids- atomic diffusion mechanism-measurement technique-range theory- implantation equipment-annealing shallow junction-high energy implantation- deposition process- chemical vapor deposition-physical vapor deposition- patterning-metallization and packaging-baked ic design. text / reference: t/r book title/author t1 s.m sze, "vlsi technology", mcgraw hill publication, 2003. t2 s.k gandhi, "vlsi fabrication principle", willy-india pvt ltd, 2008. t3 suresh babu "solid state device & technology", sanguine, 2005. r1 j.d plummer, m.d deal and peter b. griffin, "silicon vlsi technology: fundamentals, practice and modelling", pearson education publication, 2009. r2 stephen a. campbell, "fabrication engineering at the micro and nano scale", oxford university press, 2013 online resources: sl. no. website link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory- Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique- Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain 2 Understanding fabrication. (3 marks)

► Define or explain Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing. (3 marks)

► Define or explain 2 Understanding technique. (3 marks)

► Define or explain Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.. (3 marks)

Module 2

► Define or explain Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth. (3 marks)

► Define or explain 3 Understanding mechanism. (3 marks)

► Define or explain Outline the process of oxidation. (3 marks)

- **Define or explain Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.. (3 marks)**

Module 3

- **Define or explain Illustrate the concept of Lithography. (3 marks)**
- **Define or explain Explain different types of Lithographic processes. (3 marks)**
- **Define or explain Explain the concept of etching. (3 marks)**
- **Define or explain Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and. (3 marks)**

Module 4

- **Define or explain Explain Diffusion mechanism. (3 marks)**
- **Define or explain Explain various ion implantation techniques. (3 marks)**
- **Define or explain Explain different deposition mechanisms. (3 marks)**

► Define or explain Explain metallization and packaging 3**Understanding Deposition, Diffusion, Ion implantation and Packaging:**

Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow

junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and

Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1

S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K

Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3

Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D

Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication,

2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: SI.

No. Website Link 1. <http://www.asic-world.com> 2.

<https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3.

<https://www.testbench.in>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding fabrication**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain the process of Epitaxy**
3 Understanding Illustrate Epitaxial valuation and growth.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the concept of Lithography**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Explain Diffusion mechanism**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link <http://www.asic-world.com>
- <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> <https://www.testbench.in>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3371. Verify institutional updates against the current official curriculum.